GenC HR

SPE Java FSE - Angular Handbook

Why do we need this Full Stack Engineering Program?

Full Stack Engineering program engages young talents with a comprehensive learning pathway, giving these millennials an opportunity to become a Full Stack Engineer, understand the corporate environment and groom themselves even before they join us. Cognizant emphasizes on Learner Autonomy where students take charge of their own learning pathway, with the available tools and resources. More focus is given to “learning” than “teaching”. Get ready to embark your own learning adventure!

Program at a glance

Full Stack Prep-up Internship Program has 4 stages:

- Stage 1 - Core Programming Fundamentals for Full-Stack Web Development
- Stage 2 - Best Practices and Foundations of Backend Development
- Stage 3 - Backend Development and Microservices with Spring Boot and Spring Cloud
- Stage 4 - Frontend Development and Cloud Technologies
- Integrated Development Project (IDP)

Program Highlights

- The complete learning journey is formalized using adult learning principles, where problem solving and applying the skills gained are given more importance than conceptual learning.
- Learner Autonomy is implemented via Flipped Classroom for the first two stages, where the learning platform offers world class learning resources, and students would not be constrained by tutelage of an instructor.
- Higher order framework concepts would be dealt with Trainer support in Instructor Led training mode.
- Get mentored by Subject Matter Experts, whose motivation and guidance will help you accelerate in the learning journey.

Service Lines

Service lines can simply be defined as a modern organizational structure strategy for resource planning and allocation for any size of business. Typically, traditional organizational structure models are more vertically aligned -- think of an employee who has several bosses in the hierarchical ladder before being directly under the company's owner or president. Conversely, service lines follow a more horizontal continuum approach, where the company is strategically segmented into more manageable departments. The service line approach tends to focus more on the requirements of customers, which often results in noticeable increases in the customer satisfaction rate.

What is Application Development?

Application development goes through a process of planning, creating, testing, and deploying an information system, also known as the software development lifecycle. Applications are also often developed to automate some type of internal business process or processes, build a product to address common business challenges, or drive innovation.

What is Application Maintenance?

Application maintenance is the continuous updating, analyzing, modifying, and re-evaluating of your existing software applications. Application maintenance must be an ongoing task to ensure your applications are always running to the best of their abilities. Due to evolving customer expectations, the fight to survive in an existing market, and technological advancements, modifying and implementing new strategies is critical in maintaining sustainability and staying competitive. Every competitive business needs to constantly enhance and manage the IT solutions that have been developed in order to stay relevant and meet the wavering needs of users. This is where application maintenance and support come into the picture.

Contrary to popular belief, application maintenance is not just about fixing defects, but modifying a software product after delivery to correct faults, as well as to improve performance. Application maintenance and enhancement to existing applications begin with a thorough study of existing applications to identify areas of improvement.

Tips for Successfully Carrying Out Application Development and Maintenance

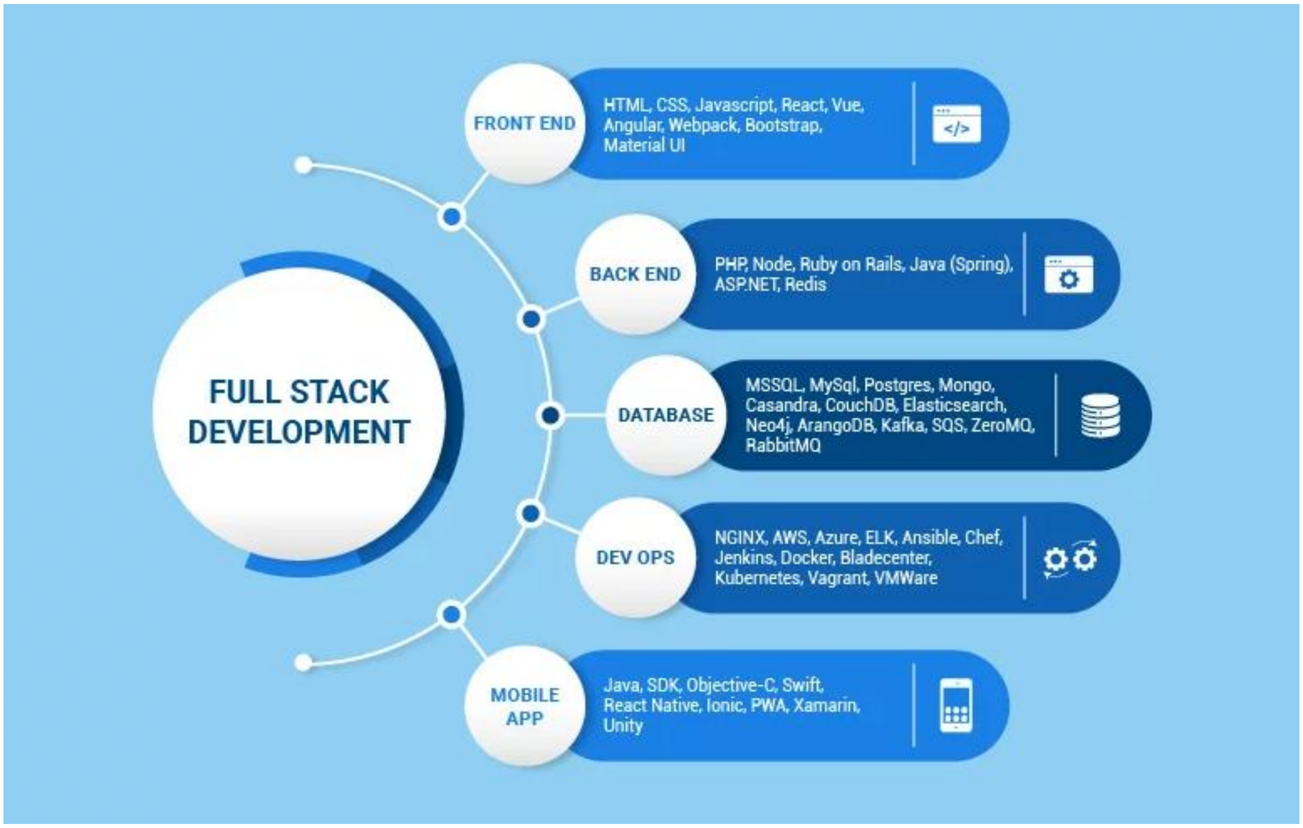
Great user experience to end customers through the development and maintenance of modern apps is a must-have. Today, applications (web or mobile) are the most cost-effective and powerful ways to reach out to a vast market and generate revenues. With millions of applications being rolled out every day, it's a good idea to keep in mind a few tips:

- Be as clear as possible as to what your requirements for your application are
- Thoroughly understand the services offered by application development companies and identify the right partner if you're using a partner
- Evaluate the various development platforms and choose the one that best fits the needs of your business

- Make sure to embed processes that focus on continuous improvements and iterations to add new features and/or fix bugs
- When developing your application, make security your top priority
- Regularly update and test your application to deliver improved and better performance, high security, and a bug-free, seamless user experience

What is Full Stack Development?

Full Stack Development (FSD) is a software development process that includes both the front and back end. To that end, a Full Stack Developer may design and create the front end while simultaneously designing, developing, and debugging databases and the software's backend. There are two significant components to full-stack application development. Development of the Front End and Back End.



Roles and Responsibilities of a Full Stack Developer

A full stack developer is responsible for both the front-end and back-end aspects of a web application. The specific roles and responsibilities can vary depending on the size of the development team and the complexity of the project, but some common responsibilities include:

1. Design and develop end-to-end web applications.
2. Implement front-end and back-end components using relevant technologies (e.g., HTML, CSS, JavaScript, React, Node.js, etc.).
3. Write clean, efficient, and well-documented code.
4. Debug and resolve technical issues.
5. Collaborate with the team and other stakeholders to deliver project on time.
6. Stay up-to-date with the latest technologies and industry trends.
7. Write automated tests to ensure code quality and improve application reliability.

8. Develop and maintain databases, servers and application deployment infrastructure.
9. Manage code repositories and version control systems (e.g., Git).
10. Participate in code reviews to ensure high-quality code.
11. Contribute to the architecture and design of applications.
12. Collaborate with designers, product managers, and other stakeholders to understand the requirements and build solutions that meet them.

Learning Journey through Flipped Classroom

This program encourages you to be more autonomous learners during out-class self-learning hours, completing the learning objectives on your own pace and style, and getting ready for the in-class practice time.

The learning path is set in the GEN C Learn Platform, which you can login with SSO.

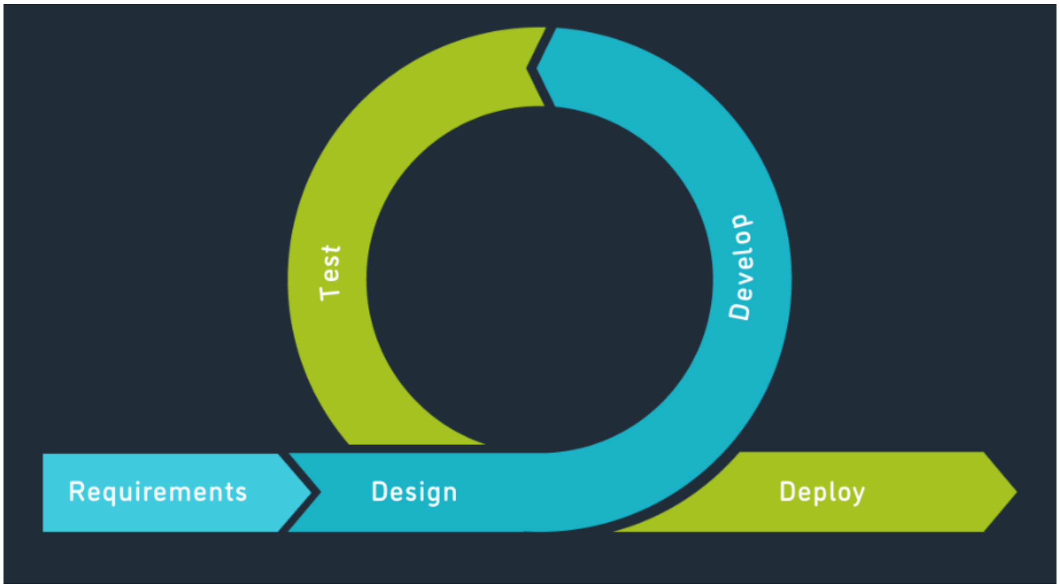
Flipped Classroom

Integrated Development Project (IDP)

What is the Integrated Development Project (IDP)?

Integrated Development Project is an approach wherein the learner experiences the entire software development processes in an incremental fashion as part of the GenC Training. The IDP implementation is purely based on **Agile Software Development** methodologies and inspired from **PBL (Project-Based Learning)** which is learning while doing. It gives learners the opportunity to gain a deeper understanding of a topic through problem-solving using real-world examples and challenges.

Following is the Agile Development Methodology at high-level.

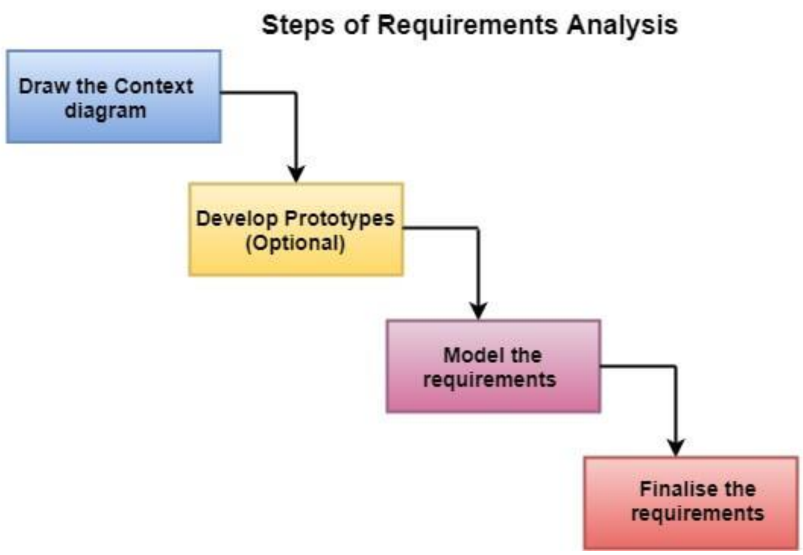


Stages of IDP

The following are the three seminal phases of IPD.

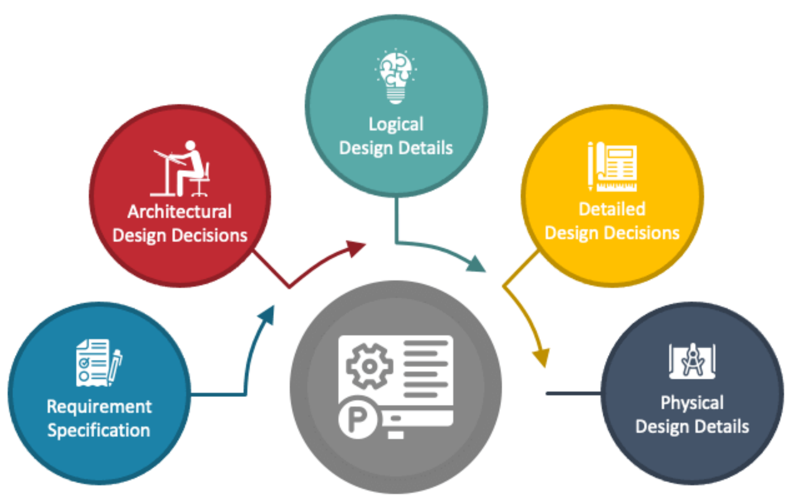
Phase 1: Requirement Analysis

Requirements analysis, also called requirements engineering, is the process of determining user expectations for a new or modified product. These features, called requirements, must be quantifiable, relevant and detailed. In software engineering, such requirements are often called functional specifications.



Phase 2: Project Design

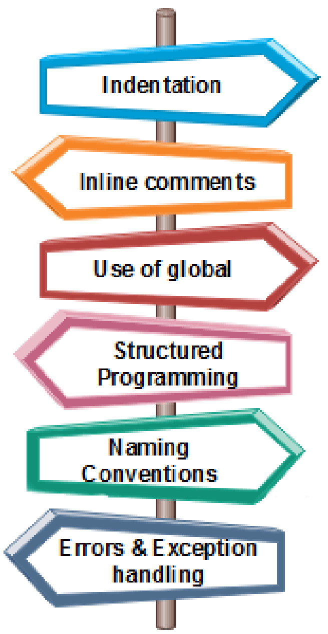
Project design is a process to transform user requirements into some suitable form, which helps the programmer in software coding and implementation.



Phase 3: Project Development

Once the system design phase is over, the next phase is development. In this phase, developers start building the entire system by writing code using the chosen programming language. In this phase, tasks are divided into units or modules and assigned to the various developers. It is the longest phase of the Software Development Life Cycle process.

Coding Standards



Recommended Program Sequence

The learning journey starts with **5 days of Icebreaker sessions, 1 day of Agile Workshop** followed by a technical learning that contains **4 stages,** and they are the following:

- **Stage 1** - Core Programming Fundamentals for Full-Stack Web Development
- **Stage 2** - Best Practices and Foundations of Backend Development
- **Stage 3** - Backend Development and Microservices with Spring Boot and Spring Cloud
- **Stage 4** - Frontend Development and Cloud Technologies
- Integrated Development Project (IDP)

Stage 1: Core Programming Fundamentals for Full-Stack Web Development

Course 1
- Agile Methodology

Course 2
- HTML5, CSS3, JavaScript
- Bootstrap 5
- jQuery

Course 3
- ANSI SQL Using MySQL

Course 4
- Core Java (v21)

Stage 1 Assessment

Stage 2: Best Practices and Foundations of Backend Development

Course 1
- SOLID Principles

Course 2
- Spring Core and Maven

Course 3
- TDD using JUnit and Mockito

Course 4
- SLF4J and Lombok

Course 5
- Code Quality Standards

Course 6
- SonarQube

Stage 3: Backend Development and Microservices with Spring Boot and Spring Cloud

Course 1
- GIT

Course 2
- Application Debugging - Backend

Course 3
- Spring Data JPA with Spring Boot, Hibernate

Course 4
- Spring REST using Spring Boot 3

Course 5
- Microservices with Spring Boot 3 and Spring Cloud

Interim Evaluation (Project + Technical)

Stage 4: Modern Frontend Development and Cloud Technologies

Course 1
- Docker Basics

Course 2
- Cloud and DevOps Basics using GCP

Course 3
- Angular (v16)

Course 4
- Application Debugging - Front-end

Final Evaluation (Project + Technical)

Final Assessment - HackerRank

There will be an integrated project called **IDP** (Integrated Development Project) which will be executed in an incremental fashion and is part of all the **4** stages.

Key Learning and Evaluation Components of the Program

Cognizant has collaborated with **Udemy** to provide world class learning videos for the evolving future of work. These Udemy programs are woven into a learning path, empowering you to plan and learn at your style.

The program also connects you with **Subject Matter Experts (SMEs)** to get professional guidance on your queries in the learning journey.

The program doesn't ONLY concentrate on the technical skilling, but also on the shaping up of the behavioral skills. **39 hours of Behavioral learning** would be done in ILT mode, with few Self-paced learning modules too.

GenCs will undergo evaluation at the end of Stage 1 of their learning in the form of a skill-based assessment. At the conclusion of Stage 3, there will be an interim evaluation during which learners will be assessed based on the technical skills covered up to that point and their progress in project deliverables. Towards the end of Stage 4, a final evaluation will take place, which will cover the entire scope of the training.

Program Completion Criteria

RAG as PHS (Performance Health Status)

The program continuously evaluates if you can apply those self-learnt skills to solve a real-time business problem. Depicted below are the two key learning components, which are distributed across the learning journey for the purpose of continuous evaluation.

Interim Evaluation:

During the interim evaluation, the GenC will undergo a video interview on the learning platform. This interview will be conducted by a Tech SME from the BU. The purpose of this evaluation is to assess the GenC's knowledge and understanding of the skills covered in the training program up to the halfway point. It also encompasses an evaluation of the GenC's progress in their Integrated Development Project (IDP). The evaluation will involve a technical discussion as well as an assessment of the IDP progression to gauge the GenC's proficiency in the skills learned thus far.

Final Evaluation:

For the final evaluation, the GenC will participate in a video interview conducted by a Tech SME from the BU. This evaluation aims to assess the GenC's knowledge and expertise in all the skills covered throughout the entire training program. Similar to the interim evaluation, this assessment will involve a technical discussion via a video interview on the learning platform, along with a project evaluation to assess the GenC's capabilities and their IDP's progress. It serves as a comprehensive evaluation of the GenC's skills and capabilities acquired during the training.

The above evaluation components will be attributed to the **Performance Health Status (PHS)** of a GenC. Additional Learning Components like Hands-On, Quizzes, CCs, and ICTs will help you to enhance your expertise level.

Mandatory Hands-On Exercise Completion

- Completion of 100% of the hands-on exercises is mandatory to qualify for stage 1 assessment, interim, and final evaluations.

Icebreaker Sessions

The icebreaker session will be conducted for a duration of an initial **5 days**. During the session, various topics related to Corporate Induction, Talent Management, Cognizant Agenda on Core Values, Leader Talks, Alumni, BU Mentor connections will be covered. Followed by icebreaker, technical training will start kicking.

The following sessions will be covered during the 5 days of icebreaker

- Corporate Induction
- Talent Manager Connect
- Cognizant Agenda Sessions on Core Values
- Leader Talks (Academy) and many more…

Learning Recommendation

A recommended day-wise schedule is provided below for the learning, with the learning content for the day, the practice hands-on and extended hands-on to be done for the day or any other activities are listed.

Stage 1: Core Programming Fundamentals for Full-Stack Web Development

Stage 1 deals with foundational technology skills that help GenCs to get start with their software engineering career. We provide unique learning experience to learners by including diversified learning content and learning methodologies that are based on adult learning principles. At the end of this stage, a **Stage 1 Assessment** will be conducted to evaluate the foundational skills of GenCs.

As part of Stage 1 of your training, the following skills will be covered.

- Agile Methodology
- HTML5, CSS3 and JavaScript
- Bootstrap, jQuery

- ANSI SQL using MYSQL
- Core Java (v21)

How and From Where to Learn?

- Udemy learnings are recommended in the Platform to understand the fundamental concepts. In addition to this, you can also learn from any other sources as they are mentioned in this handbook.

Stage 1 -> Course 1 -> Agile Methodology

Course Overview

In **Course 1** of **Stage 1**, learners will be introduced to the fundamentals of **Agile methodology**. Agile is a project management and software development approach that emphasizes flexibility, collaboration, and customer satisfaction. It promotes adaptive planning, iterative development, early delivery, and continuous improvement. Agile methodologies, such as Scrum and Kanban, focus on delivering value to the customer and effectively responding to change.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand the principles and values of Agile methodology.
- Describe the benefits of using Agile in software development.
- Explain the differences between Agile and traditional project management approaches.
- Identify the key roles and responsibilities in Agile teams.
- Describe the iterative and incremental nature of Agile development.
- Explain the importance of customer collaboration and feedback in Agile.
- Describe common Agile practices, such as user stories, sprints, and retrospectives.
- Identify common Agile frameworks, such as Scrum, Kanban, and Extreme Programming (XP).
- Explain how Agile principles can be applied in different project environments.

Day 1

Agile Methodology

Key Topics: Agile Basics

Continuous Learning: Technical Enablement

Learn about Agile process from the Udemy course below.

Agile Crash Course: Agile Project Management; Agile Delivery

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **All** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Stage 1 -> Course 2 -> HTML5, CSS3, JavaScript, Bootstrap5, jQuery

Course Overview

Course 2 of **Stage 1** focuses on Web UI technologies, including **HTML5, CSS3, JavaScript, Bootstrap 5,** and **jQuery**. These technologies are fundamental for building modern, responsive, and interactive user interfaces for web applications. HTML5 provides the structural foundation of web pages, CSS3 enables styling and layout customization, and JavaScript adds interactivity and dynamic behavior. Bootstrap 5 is a widely used CSS framework that simplifies the creation of responsive designs, while jQuery is a JavaScript library that streamlines HTML document traversal, manipulation, event handling, and animation. Mastering these technologies is essential for developing engaging and user-friendly web applications.

Learning Objectives

After completing this course, GenCs will be able to:

- Define HTML and common terminology related to HTML, recognize correct HTML syntax, and write a brief error-free HTML code.
- Apply style to an existing/new web page as per the requirement using CSS3.
- Write and employ JavaScript code to solve practical web design problems.
- Make responsive, cross-platform and modern websites using Bootstrap4.
- Illustrate animated, interactive web pages using jQuery libraries.

Day 2

HTML5, CSS3

Key Topics: HTML5 - Introduction, Getting Started, Elements & Attributes, Navigation, Events, Web Forms 2.0, Web Storage, Web SQL Database, Geo location; CSS3 - Introduction, Selectors, Styling, Box Model, Advanced

Continuous Learning: Technical Enablement

The HTML & CSS Bootcamp 2025 Edition

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Introduction
    - **Section 2:** HTML Basics
    - **Section 3:** More HTML
    - **Section 4:** Working With Forms
    - **Section 5:** Other Elements
    - **Section 6:** CSS Basics
    - **Section 7:** The World of CSS Colors
    - **Section 8:** Styling Text
    - **Section 9:** More Text Styling
    - **Section 10:** Selectors Pt. 1
    - **Section 11:** The Box Model
    - **Section 15:** CSS Units
    - **Section 17:** Backgrounds, Gradients, & Filters
    - **Section 18:** Positioning
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Newberry Library
- Jasper Sports Academy
- WhatsApp Chat
- Calculator
- 4*4 Sudoku

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level in HTML5 and CSS3.

- HTML 5, CSS3

Code Challenge (For Practice Only)

Attempt the following Code Challenge through the Learning Path at Tekstac for checking your skill level in HTML5 and CSS3. You must secure 70% to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - HTML5 and CSS3

Day 3

JavaScript

Key Topics: Getting Started, Fundamentals, Operators, Control Flow, Objects, Arrays

Continuous Learning: Technical Enablement

Javascript basics for beginners

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Getting Started
    - **Section 2:** Basics
    - **Section 3:** Operators
    - **Section 4:** Control flow
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Schedule Upcoming Match
- Event Reservation Portal
- Display Student Details

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac to check your knowledge level in JavaScript.

- Java Script

Additional Learning

Go through the Udemy course to understand the usage of Chrome Developer Tools, a comprehensive toolkit for developers built directly into the Chrome browser. These tools allow you to edit web pages in real-time, diagnose problems more quickly, and build better websites faster.

Devtools Pro: The Basics of Chrome Developer Tools

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **All** Sections

- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 4 - Forenoon

JavaScript

Key Topics: JavaScript Deep Dive

Continuous Learning: Technical Enablement

Javascript basics for beginners

- Learn the sections listed below in this Udemy course and complete the corresponding hands-on coding given below.
  - **Section 5:** Objects
  - **Section 6:** Arrays
  - **Section 7:** Functions
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Counting Game
- SignUp Form validation
- Fitness Centre
- Product Code Validation

Code Challenge (For Practice Only)

Attempt the following Code Challenges through the Learning Path at Tekstac for checking your skill level on JavaScript. There will be only 3 attempts, and you must secure 70% to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - JavaScript

Course Overview

Course 3 of **Stage 1** provides a comprehensive introduction to **Bootstrap 5**, the world's most popular front-end framework for building responsive, mobile-first websites. GenCs will explore the fundamental concepts and tools that Bootstrap 5 offers, empowering them to design and develop modern, dynamic web interfaces efficiently. From understanding the layout and grid system to leveraging essential components, utilities, and helpers, this course will equip them with the skills to create visually appealing and user-friendly web pages. Additionally, GenCs will learn how to customize and extend Bootstrap to meet specific project requirements while mastering responsive design principles.

Learning Objectives

After completing this course, GenCs will be able to:

- Describe the purpose and features of Bootstrap 5 as a front-end framework.
- Design responsive web layouts using the Bootstrap grid system.
- Implement advanced layout techniques for complex web pages.
- Incorporate common UI elements like navigation bars, modals, buttons, and cards to enhance interactivity and usability.
- Use Bootstrap utilities and helper classes to streamline the styling and functionality of web pages.
- Develop web pages that adapt seamlessly to different devices and screen sizes.
- Leverage responsive breakpoints effectively for design adjustments.
- Modify default themes and styles to align with specific design requirements.
- Integrate Bootstrap with custom CSS and JavaScript to extend its functionality.

Day 4 - Afternoon

Bootstrap 5

Key Topics: Introduction to Bootstrap 5, Layout and Grid System

Continuous Learning: Technical Enablement

The Ultimate Bootstrap Guide - Bootstrap 5 from Scratch

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 1:** Getting Started with Bootstrap 5
  - **Section 2:** Layouts in Bootstrap 5
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Ohm's Law

Day 5 - Forenoon

Bootstrap 5

Key Topics: Essential Components, Utilities and Helpers, Responsive Design, Customization and Extensions

Continuous Learning: Technical Enablement

The Ultimate Bootstrap Guide - Bootstrap 5 from Scratch

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 5:** Components in Bootstrap 5
    - **Section 6:** Helpers in Bootstrap 5
    - **Section 7:** Utilities in Bootstrap 5
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Learning References

- Responsive Web Design - Media Queries
- Responsive Web Design - Images
- Bootstrap 5 Navbar Responsive behaviors Toggler
- How to theme, customize, and extend Bootstrap with Sass

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Sticky Notes
- Portfolio
- Library Home Page
- Hi-Tech Digital world

Course Overview

Course 4 of **Stage 1** is designed to provide a comprehensive introduction to jQuery, a powerful JavaScript library that simplifies web development by streamlining tasks such as DOM manipulation, event handling, and AJAX integration. Through practical examples, GenCs will learn how to leverage jQuery's robust features to enhance user interfaces and create dynamic, interactive web applications.

Learning Objectives

After completing this course, GenCs will be able to:

- Explain the purpose and benefits of using jQuery in web development.
- Identify the core features and components of the jQuery library.
- Select and manipulate HTML elements efficiently using jQuery selectors and methods.
- Apply various jQuery methods to alter content, attributes, and styles dynamically.
- Perform advanced DOM manipulation, including traversing and filtering elements.
- Attach and manage event handlers to create interactive web experiences.
- Use jQuery's AJAX capabilities to send and receive data asynchronously.

Day 5 - Afternoon

jQuery

Key Topics: jQuery and its features, Basic components, DOM manipulation & events, Basic AJAX with jQuery

Continuous Learning: Technical Enablement

The Complete jQuery Course: From Beginner To Advanced!

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Introduction
    - **Section 3:** Element Selectors
    - **Section 4:** Manipulating the DOM I – Inserting, Replacing and Removing Elements
    - **Section 5:** Manipulating the DOM II – Changing Element Data and CSS
    - **Section 6:** Events I – Handling Mouse Events & Keyboard Events
    - **Section 7:** Events II – Forms
    - **Section 8:** Ajax with jQuery
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Course Overview

In **Course 5** of **Stage 1**, the focus will be on **ANSI SQL using MySQL**. ANSI SQL (American National Standards Institute Structured Query Language) is a standardized version of SQL that ensures compatibility across different database systems. MySQL is a popular open-source relational database management system that uses SQL for database access. This milestone will cover key concepts of ANSI SQL, such as querying databases, modifying data, creating and managing database objects, and implementing data integrity. By using MySQL as the implementation platform, GenCs will gain practical experience in applying ANSI SQL concepts to a real-world database system, preparing them for database development tasks in various IT roles.

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Learning Objectives

After completing this course, GenCs will be able to:

- Learn ANSI SQL fundamentals and their application in MySQL.
- Practice retrieving and manipulating data using the SELECT statement in MySQL.
- Apply filtering, sorting, and grouping techniques to query data effectively.
- Gain an understanding of join operations and subqueries for combining data from multiple tables.
- Develop skills in modifying data with INSERT, UPDATE, and DELETE statements in MySQL.

Day 6

ANSI SQL Using MySQL

Key Topics: Introduction to ANSI SQL and MySQL, Data Retrieval with SELECT Statement, Filtering and Sorting Data, Aggregate Functions and Grouping

Continuous Learning: Technical Enablement

SQL for Beginners: Learn SQL using MySQL and Database Design

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 2:** Installation and Setup
    - **Section 6:** Selecting from a Table
    - **Section 10:** Aggregate Functions
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the solution. Write the query yourself.

- Diverse Selection
- Email Providers
- Book Event
- Upcoming Fair
- Routine Checks
- Security Measure
- May Birthday
- Age Demographics
- Author Detail
- Price Analysis
- Updated Email
- Lended Books
- The Chronicles of Authors and Their Tales

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level on querying database.

- Querying Database - Operators, Aggregate, String, Date Functions

Day 7, Day 8 - Forenoon

ANSI SQL Using MySQL

Key Topics: Joins and Subqueries, Data Modification with INSERT, UPDATE, DELETE, Creating and Modifying Tables, Indexes and Constraints

Continuous Learning: Technical Enablement

SQL for Beginners: Learn SQL using MySQL and Database Design

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 7:** Selecting From Multiple Tables
  - **Section 8:** Database Design
  - **Section 11:** Subqueries
  - **Section 3:** Data Definition Language
  - **Section 4:** More On Alter Table
  - **Section 5:** Data Manipulation Language
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the solution. Write the query yourself.

- Date Format
- Exploring Users' Literary Journeys
- Active Borrowings with User and Book Details
- Same Authors
- User Borrowing Summary
- Maximum Fine Per User
- Top Users
- User Count by Title and Author
- Create Borrowing Table
- Alter Table User
- Alter Table Book
- Remove Table
- Insert User Table
- Update Record
- Delete Record

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level on SQL DDL and DML statements.

- SQL basics with DML and DDL Statements

Code Challenge (For Practice Only)

Attempt the following Code Challenges through the Learning Path at Tekstac for checking your skill level on ANSI SQL. There will be only 3 attempts, and you must secure 70% to clear this challenge.

Do not copy paste the solution. Write the query yourself.

- Assess-Type-1: Code Challenge - RDBMS Select Statements
- Assess-Type-1: Code Challenge - RDBMS DDL & DML
- Assess-Type-1: Code Challenge - Function, Scalar & Aggregate
- Assess-Type-1: Code Challenge - Functions & SubQueries

Course Overview

Course 6 of **Stage 1** offers a comprehensive introduction to **Java programming**, designed for learners aiming to build a strong foundation in one of the most widely used programming languages. Starting with core concepts, the course progressively delves into advanced topics such as Object-Oriented Programming (OOP), modern Java features, and reactive programming. By the end of the course, learners will be equipped to develop efficient, scalable, and maintainable applications while understanding key Java frameworks and tools essential for enterprise-level programming.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand the basics of Java, including its syntax, data types, and program structure.
- Develop Java programs using standard coding practices and conventions.
- Apply core OOP principles such as encapsulation, inheritance, polymorphism, and abstraction in Java.
- Implement abstract classes, interfaces, and annotations to design robust and flexible programs.
- Utilize essential classes and utilities from the java.lang package.
- Effectively manage data using the Java Collections Framework.
- Handle file operations and perform efficient Input/Output (I/O) tasks.
- Implement exception handling mechanisms to create error-resilient applications.
- Explore modern Java features, including Streams, Lambda Expressions, and enhancements in Java 17 and 21.
- Manage date and time operations using the java.time package.
- Develop multithreaded programs and understand concurrency mechanisms for high-performance applications.
- Build networking applications using sockets and the Java Networking API.
- Connect and interact with relational databases using JDBC.
- Understand reactive programming principles and their benefits in modern application development.
- Build scalable and responsive applications using Java Reactive Streams and frameworks.
- Learn basic reverse engineering concepts and enhance debugging skills.

Day 8 - Afternoon, 9

Core Java (v21)

Key Topics: Introduction to Java, Basics of Java Programming

Continuous Learning: Technical Enablement

Java Programming For Beginners | Core Java Using IntelliJ

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 20:** [NEW] Introduction And Setup - The Java Environment
  - **Section 21:** [NEW] Introduction to Java Programming - The First Steps
  - **Section 22:** [NEW] Variables and Datatypes in Java
  - **Section 23:** [NEW] Expressions and Operators in Java
  - **Section 26:** [NEW] Interactive Programming with Java
  - **Section 29:** [NEW] Arrays
  - **Section 30:** [NEW] Multi Dimensional Arrays
  - **Section 31:** [NEW] Strings and StringBuffer
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Streamline Your Health Journey
- Water Tank Capacity
- Square Series
- Automate Sum Finder
- Number Crunch Challenge
- Middle Number Game
- WordZee
- Fishing competition

Technical Quiz

Attempt the following technical quiz in the Learning Path at Tekstac for checking your knowledge level on Java fundamental concept.

- Java Operator, Control flow statement

Day 10

Core Java (v21)

Key Topics: Object-Oriented Programming in Java

Continuous Learning: Technical Enablement

Java Programming For Beginners | Core Java Using IntelliJ

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 32:** [NEW] OOPS: Classes and Objects | Object Oriented Programming In Java
    - **Section 34:** [NEW] Modifiers in Java
    - **Section 35:** [NEW] OOPS: Inheritance | Object Oriented Programming In Java
    - **Section 36:** [NEW] OOPS: Polymorphism | Object Oriented Programming In Java
    - **Section 37:** [NEW] OOPS: Abstraction | Object Oriented Programming In Java
    - **Section 38:** [NEW] OOPS: Encapsulation | Object Oriented Programming In Java
    - **Section 39:** [NEW] Nested Classes in Java
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Brilliant Restaurant
- Investment Calculation
- Dazzle Closet
- Mass Of Rocket Component
- Lego Builders
- Plan Smart
- Mythic Rhythms
- Fitness Tracker

Day 11

Core Java (v21)

Key Topics: Exception Handling, Java Collections Framework

Continuous Learning: Technical Enablement

Java Programming For Beginners | Core Java Using IntelliJ

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 14:** Collections in Java
    - **Section 43:** [NEW] Generics in Java
    - **Section 40:** [NEW] Exception handling in Java
    - **Section 41:** [NEW] Custom Exceptions in Java

- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- College Namelist
- Babitha's App
- Alta Motors
- Concept And Implementation
- Student Score Info
- Vintage Books Emporium
- Animalia
- Array Aggregate
- Product Details
- Parse the Dates

Day 12

Core Java (v21)

Key Topics: Functional Programming in Java

Continuous Learning: Technical Enablement

Java Programming For Beginners | Core Java Using IntelliJ

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 44:** [NEW] Functional Interfaces and Lambda expressions
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- The Perfect Password
- Area And Volume
- Mystic Line

Day 13

Core Java (v21)

Key Topics: Java I/O and File Handling

Continuous Learning: Technical Enablement

Java Programming For Beginners | Core Java Using IntelliJ

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 115:** File handling using Java IO API
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

- Copy Names

Day 14

Core Java (v21)

Key Topics: Multithreading and Concurrency

Continuous Learning: Technical Enablement

Java Programming For Beginners | Core Java Using IntelliJ

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 17:** Concurrency in java
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

- Library Management

Day 15

Core Java (v21)

Key Topics: Application Debugging Using IntelliJ IDEA, Introduction to Reactive Programming

Continuous Learning: Technical Enablement

Java Debugging With IntelliJ IDEA

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **ALL** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Reactive Programming in Modern Java using Project Reactor

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **ALL** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 16

Core Java (v21)

Key Topics: Working with Java Modules, Java Networking, Reverse Engineering Concepts

Continuous Learning: Technical Enablement

Java Programming For Beginners | Core Java Using IntelliJ

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 16:** Java Platform Module System
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Java Network Programming - TCP/IP Socket Programming

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 1:** Introduction to Networking
  - **Section 4:** Java Sockets
  - **Section 5:** Java UDP programming

- **Section 8:** Basic Web Server Application
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Reverse Engineering and Malware Analysis Fundamentals

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **ALL** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 17, 18 - Forenoon

Core Java (v21)

Key Topics: Language-Specific Features - v17, v21, Introduction to Java Database Connectivity (JDBC)

Continuous Learning: Technical Enablement

Java 21: New Features and Enhancements - 2024

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **ALL** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Java Database Connection: JDBC and MySQL

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **ALL** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

- Customer Record
- Fit Freak
- Employee Details
- Booking System

Day 18 - Afternoon

- This part of the day has been added to adjust the behavioral training and mentor connect.

Stage 1 Assessment

Day 19

Mock Assessment, Stage 1 Assessment

- This day will be dedicated for the Stage 1 Assessment and Result Publishing

Assessment Overview

- **Assessment Type**: Skill-Based Assessment (SBA) + Knowledge-Based Assessment
- **Duration**: 180 minutes
- **Number of Questions**: 17 (SBA - Core Java - 3, SQL - 2, KBA - Web UI - 12)
- **Modules Covered**:
  - Core Java
  - SQL
  - Web UI (HTML5, CSS3, JavaScript)
- **Maximum Attempts**: 3

Stage 2 -> Course 1 -> SOLID Principles

Course Overview

Course 1 of **Stage 2** introduces you to the foundational principles of Object-Oriented Programming (OOP) through the lens of the SOLID principles. These principles, widely regarded as best practices in software design, ensure robust, maintainable, and scalable code. By understanding "What" these principles entail and "Why" they are essential, you will develop the skills necessary to design and implement software that adheres to high standards of quality and efficiency.

Through an exploration of the Single Responsibility Principle, the Open-Closed Principle, the Liskov Substitution Principle, the Interface Segregation Principle, and the Dependency Inversion Principle, you will learn how to create solutions that are flexible and resistant to design flaws.

Learning Objectives

After completing this course, GenCs will be able to:

- Explain what the SOLID principles are and why they are critical for effective software design.
- Design classes and modules that have a single, clear purpose to improve code readability and maintainability.
- Write code that can be extended without modifying the existing implementation, ensuring adaptability to future requirements.
- Ensure subclasses can replace their parent classes without disrupting the application's functionality.
- Create interfaces that are specific to the needs of clients, avoiding the pitfalls of bloated or unnecessary functionalities.
- Invert dependencies to rely on abstractions rather than concrete implementations, promoting decoupled and testable code.
- Integrate SOLID principles into your development process to create maintainable and scalable applications.

Day 20

SOLID Principles

Key Topics: SOLID Principles

Continuous Learning: Technical Enablement

Design Patterns in Java

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 2:** SOLID Design Principles
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the hands-on given in the Udemy course while you learn.

Do not copy paste the code. Write the code yourself.

Stage 2 -> Course 2 -> Spring Core and Maven

Course Overview

Course 2 of **Stage 2** is designed to equip learners with a comprehensive understanding of **Spring Framework** fundamentals and the **Maven** build tool. This course provides hands-on experience in developing robust, scalable, and maintainable Java-based applications using the Spring Framework and Maven. Learners will explore key Spring concepts like Inversion of Control (IoC), Dependency

Injection (DI), Aspect-Oriented Programming (AOP), and Spring MVC. Additionally, they will gain foundational knowledge of Maven, including setting up builds, managing dependencies, and using plugins to streamline project workflows. The course concludes with an introduction to advanced topics such as Spring Boot and Reactive Programming with Spring WebFlux, preparing learners for modern application development challenges.

Learning Objectives

After completing this course, GenCs will be able to:

- Describe the core concepts and architecture of the Spring Framework.
- Set up and configure Spring projects using Maven.
- Explain the role and functionality of the Spring IoC Container.
- Configure Spring Beans and manage their lifecycle.
- Use Dependency Injection (DI) to decouple application components.
- Apply DI for improved application flexibility and testability.
- Understand the principles of Aspect-Oriented Programming (AOP).
- Implement cross-cutting concerns like logging and security using Spring AOP.
- Build web applications using Spring MVC.
- Integrate Spring with Object-Relational Mapping (ORM) tools.
- Understand the purpose and benefits of Spring Boot.
- Explore the basics of Spring Boot project setup and configurations.
- Understand the principles of Reactive Programming.
- Create reactive web applications using Spring WebFlux.
- Understand Maven's role in project management and build automation.
- Work with Maven’s build lifecycle, manage dependencies, and use plugins effectively.
- Execute Maven builds and troubleshoot common issues.

Day 21

Spring Core and Maven

Key Topics: Introduction to Spring Framework, Spring IoC Container, Spring Bean Configuration

Continuous Learning: Technical Enablement

Spring Framework In Easy Steps

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 2:** Software Setup
    - **Section 3:** Spring Core Concepts
    - **Section 5:** Life Cycle Methods
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Display Staff Details
- PatientManagement

Day 22

Spring Core and Maven

Key Topics: Dependency Injection in Spring, Spring AOP (Aspect-Oriented Programming)

Continuous Learning: Technical Enablement

Spring Framework In Easy Steps

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 4:** Setter Injection
  - **Section 6:** Dependency Check, Inner beans and Scopes
  - **Section 7:** Constructor Injection
  - **Section 9:** Auto-Wiring
  - **Section 22:** Spring AOP
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Display Book Details - Autowiring
- Gold Rate Calculation - Collections
- Shipment-Item-Scope
- Account-Loan
- DBConfig-SetterBasedInjection
- Loan Pro
- Azure Horizon Resort
- Hospital Management
- Smart Home Lighting System
- SpringAopDemo

Day 23

Spring Core and Maven

Key Topics: Spring MVC and ORM, Spring Boot (Introduction)

Continuous Learning: Technical Enablement

Spring Framework In Easy Steps

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 20:** Spring MVC and ORM
    - **Section 25:** Spring Boot
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Billing Software Application
- EBill

Day 24

Spring Core and Maven

Key Topics: Reactive Programming with Spring WebFlux

Continuous Learning: Technical Enablement

Reactive Programming with Spring Framework 5

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 2:** Introduction to Reactive Programming
    - **Section 3:** Netflux Reactive Example
    - **Section 4:** Spring Web Client
    - **Section 5:** Spring WebFlux
    - **Section 6:** R2DBC
    - **Section 7:** Functional Endpoints with Spring WebFlux
    - **Section 8:** Reactive Streams
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 25 - Forenoon

Spring Core and Maven

Key Topics: Maven Basics, Build Lifecycle, Plugins, Running Maven Builds

Continuous Learning: Technical Enablement

Maven Crash Course

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 1:** Introduction
  - **Section 2:** Simple Software Setup
  - **Section 3:** Maven Project Creation and Key Concepts
  - **Section 4:** Maven in Eclipse
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Build Web Application
- Compiling Executing Java using POM
- Junit With Maven
- Maven Directory Structure
- Maven Shade Plugin
- Maven App to access External Service
- Capital Service

Code Challenge (For Practice Only)

Attempt the following Code Challenge through the Learning Path for checking your skill level on Spring Framework. There will be only 3 attempts, and you must acquire 70% to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - Spring Framework

Course Overview

Course 3 of **Stage 2** provides a comprehensive introduction to modern testing practices essential for building high-quality software. It is designed to equip learners with the skills and knowledge to implement **Test-Driven Development (TDD)** effectively and to use powerful testing frameworks such as **JUnit** and **Mockito**. Through hands-on practice, learners will gain expertise in testing Java applications, leveraging advanced features of JUnit, and creating mock objects to isolate dependencies. Special focus is placed on testing Spring-based applications and managing external dependencies to ensure robust, maintainable, and scalable software solutions.

Learning Objectives

After completing this course, GenCs will be able to:

- Grasp the principles, benefits, and workflow of TDD in the software development lifecycle.
- Set up and use JUnit for writing and executing unit tests in Java applications.
- Leverage advanced JUnit features to create more expressive and efficient tests.
- Understand the fundamentals of Mockito for mocking objects and isolating dependencies in unit testing.
- Apply JUnit and Mockito to test Spring-based applications, ensuring the reliability of components and configurations.
- Implement effective mocking strategies to simulate external dependencies and test the application in isolation.

Day 25 - Afternoon

TDD using JUnit and Mockito

Key Topics: Introduction to Test-Driven Development (TDD), Getting Started with JUnit, Advanced JUnit Features

Continuous Learning: Technical Enablement

Learn Java Unit Testing with Junit & Mockito in 30 Steps

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 2:** Unit Testing with JUnit
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- EB Connection
- CarcoExpress

- Employee Details
- Employee Appraisal

Day 26, 27 - Forenoon

TDD using JUnit and Mockito

Key Topics: Mockito Basics, Testing Spring Applications with JUnit and Mockito, Mocking External Dependencies

Continuous Learning: Technical Enablement

Spring Boot Unit Testing with JUnit, Mockito and MockMvc

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 5:** Unit Testing - Mocking with Mockito
  - **Section 7:** Testing Spring Boot MVC Web Apps - Database Integration Testing
  - **Section 8:** Testing Spring Boot MVC Web Apps - MVC Controller Testing
  - **Section 12:** Testing Spring Boot REST APIs
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Employee Address Update
- Computation Service
- Employee Details (Mockito)
- Crisp Placement Training

Code Challenges (For Practice Only)

Attempt the following Code Challenges through the Learning Path for checking your skill level on TDD and JUnit. There will be only 3 attempts and you have to score 70% in order to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - TDD, Junits

Course Overview

Course 5 of **Stage 2** provides an in-depth understanding of **SLF4J** and **Lombok**, two powerful libraries that simplify and enhance Java development. You will explore the logging capabilities of SLF4J and its role as a façade for various logging frameworks, along with practical techniques for efficient logging and integration with popular tools. Additionally, the course delves into Lombok's ability to reduce boilerplate code through annotation-based code generation, covering essential features such as getter and setter methods, constructors, and builder patterns. The course equips learners with the skills to streamline code, improve maintainability, and leverage these tools effectively in real-world projects.

Learning Objectives

After completing this course, GenCs will be able to:

- Explain the key features and benefits of SLF4J as a logging façade.
- Compare SLF4J with other logging frameworks, including Log4j and Java Util Logging (JUL).
- Demonstrate how to add and manage SLF4J dependencies using Maven and Gradle.
- Integrate SLF4J with existing logging frameworks and popular libraries like Logback.
- Utilize the Logger interface and its methods to implement effective logging practices.
- Apply best practices for parameterized logging and avoid performance pitfalls such as string concatenation.
- Describe the benefits of Lombok in reducing boilerplate code and improving developer productivity.
- Integrate Lombok seamlessly with popular IDEs like IntelliJ IDEA and Eclipse.
- Leverage Lombok annotations such as @Getter, @Setter, @ToString, and @EqualsAndHashCode for efficient code generation.
- Construct classes effectively using annotations like @NoArgsConstructor, @AllArgsConstructor, @RequiredArgsConstructor, and @Builder.
- Utilize advanced annotations like @Slf4j for logging, @Value for immutability, and @With for immutability-based methods.
- Configure and troubleshoot Lombok in build tools (Maven, Gradle) and resolve common annotation processing issues.
- Debug and analyze Lombok-generated code to ensure alignment with project requirements.

Day 27 - Afternoon, 28 - Forenoon

SLF4J and Lombok

Key Topics: Introduction to SLF4J, Setting Up SLF4J, Core Concepts of SLF4J, Introduction to Lombok, Key Lombok Annotations, Advanced Lombok Features, Lombok and IDEs

Continuous Learning: Technical Enablement

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 23:** Logging in Java
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Learning Reference

Practice the sample programs and code snippets provided in the following links.

- Lombok @Log4j, @Slf4j and Other Log Annotations

Stage 2 -> Course 5 -> Code Quality Standards

Course Overview

Course 5 of **Stage 2** is designed to provide a comprehensive understanding of code quality and its significance in modern software development. GenCs will explore best practices, industry standards, and tools to ensure clean, maintainable, and secure code. The curriculum focuses on fostering discipline in coding practices, improving team collaboration through effective code reviews, and leveraging tools to measure and maintain high-quality code standards. By integrating static code analysis, peer review techniques, and industry-approved metrics, learners will gain hands-on experience in writing and evaluating code that meets professional benchmarks.

Learning Objectives

After completing this course, GenCs will be able to:

- Explain the significance of maintaining high standards in code quality.
- Identify the benefits and common challenges in achieving quality assurance in code.
- Implement naming conventions, formatting standards, and effective documentation.
- Organize code using best practices for package and import statements.
- Gain proficiency in configuring and running tools like Checkstyle, PMD, and SpotBugs.
- Analyze and interpret results from static code analysis tools to improve code quality.
- Conduct thorough and constructive code reviews using tools such as GitHub Pull Requests, Gerrit, and Crucible.
- Promote team collaboration by following best practices for peer reviews.
- Calculate and interpret key metrics such as cyclomatic complexity, code duplication, and maintainability index.
- Use tools like SonarQube and IntelliJ IDEA to measure and monitor code quality, and set appropriate thresholds for quality assurance.
- Integrate OWASP Secure Coding Guidelines for Java into development practices.
- Align with ISO/IEC standards for code quality to ensure compliance with industry

benchmarks.

Day 28 - Afternoon

Code Quality Standards

Key Topics: Introduction to Code Quality Standards, Java Coding Conventions, Static Code Analysis Tools, Code Reviews, Code Quality Metrics, Compliance and Industry Standards

Continuous Learning: Technical Enablement

Java Best Practices for Efficient, Scalable, and Secure Code

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 21:** Best Practices of Secure Coding in Java
  - **Section 25:** Best Practices of REST Architecture in Java Applications
  - **Section 26:** Best Practices of Tracking Software Development Performance, Clean Code & others
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Learning Reference

- What is Code Quality?
- Naming Conventions in Java
- Java Static Analysis Tools in Eclipse and IntelliJ IDEA
- 30+ Java Code Review Checklist Items
- Continuous Code Quality using SonarQube

Stage 2 -> Course 6 -> SonarQube

Course Overview

Course 6 of **Stage 2** offers a comprehensive introduction to **SonarQube**, an industry-leading platform for continuous code quality inspection. Designed for developers, architects, and DevOps professionals, the course delves into the critical role of SonarQube in identifying technical debt and improving software maintainability, reliability, and security. Participants will explore the tool's architecture, installation methods, and key functionalities such as setting up quality gates, scanning codebases, and integrating SonarQube with CI/CD pipelines and popular development tools. By the end of the course, learners will be equipped to effectively use SonarQube to enhance code quality and streamline the development process.

Learning Objectives

After completing this course, GenCs will be able to:

- Define SonarQube, its purpose, and the benefits of continuous code quality inspection.
- Identify and address technical debt using SonarQube.

- Recognize the supported programming languages and the platform's architectural components.
- Perform on-premises and Docker-based installations of SonarQube.
- Configure databases and initiate the SonarQube server.
- Access and navigate the SonarQube web interface.
- Manage projects and dashboards within SonarQube.
- Configure quality profiles and gates to enforce coding standards.
- Detect and resolve issues, code smells, and technical debt.
- Utilize the SonarQube scanner to analyze codebases.
- Integrate SonarQube with CI/CD pipelines, build tools, and IDEs.
- Measure and interpret code coverage, duplication, maintainability, reliability, and security metrics.
- Generate detailed reports to track and communicate code quality improvements.

Day 29

SonarQube

Key Topics: Introduction to SonarQube, Installing and Setting Up SonarQube, Key Concepts and Features, Integrating SonarQube into Development, Metrics and Reporting

Continuous Learning: Technical Enablement

Continuous Code Inspection with SonarQube

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **ALL** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Walk through the following videos to practice SonarQube.

- SonarQube - Part 1
- SonarQube - Part 2
- SonarQube - Part 3
- SonarQube - Part 4
- SonarQube - Part 5
- SonarQube - Part 6

Code Challenges (For Practice Only)

Attempt the following Code Challenges through the Learning Path for checking your skill level on Code Quality. There will be only 3 attempts and you have to score 70% in order to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - Code Quality

ICT (Integrated Capability Test) (For Practice Only)

Take up the following extended integrated practice task to check your skill level after completing the Stage 2 of your training. Unlike Code Challenge, the coverage of this practice will be Spring MVC Spring Boot and ORM. There will be only 3 attempts, and you must score a minimum 70% to complete this activity successfully.

Do not copy paste the code. Write the code yourself.

- Java Assess-Type-2: Integrated Capability Test (ICT)

IDP - Project Activities

Day 30

Internal Demo and Rework

- This day can be utilized for Internal Demo & Re-work

Stage 3 - Backend Development and Microservices with Spring Boot and Spring Cloud

Overview

Stage 3 of the training delves into advanced backend development skills with a focus on **Java and Spring Boot**. GenCs will learn about using **GIT** for version control to manage source code effectively. They will also gain expertise in debugging backend applications, learning techniques to identify and resolve issues efficiently. The training covers **Spring Data JPA with Spring Boot and Hibernate**, enabling GenCs to interact with relational databases seamlessly. Additionally, they will master building RESTful web services using **Spring Boot 3**, a framework highly regarded for Java-based applications. The stage culminates with a deep dive into microservices architecture, where GenCs will learn to design and implement microservices using **Spring Boot 3 and Spring Cloud**, providing them with the necessary tools for building and deploying microservices-based applications.

As part of the **Stage 3** of your training, the following skills will be covered.

- GIT
- Application Debugging - Backend
- Spring Data JPA with Spring Boot, Hibernate
- Spring REST using Spring Boot 3
- Microservices with Spring Boot 3 and Spring Cloud

How and From Where to Learn?

- You can learn from the sources as they are mentioned in this learning guide.

Stage 3 -> Course 1 -> GIT

Course Overview

In **Course 1** of **Stage 3**, you will dive into the fundamental skill of version control using **Git**. Git is a distributed version control system that allows you to track changes to your codebase, collaborate with other developers, and manage different versions of your project. You will learn the basics of Git, including setting up a repository, committing changes, branching and merging, resolving conflicts, and working with remote repositories. Understanding Git is essential for any developer, as it provides a solid foundation for managing and collaborating on projects effectively.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand the concept of version control and its importance in managing project files and code changes efficiently.
- Gain a comprehensive understanding of Git, including its features, advantages, and how it differs from other version control systems.
- Learn how to set up Git on your local machine, including configuring user details and initializing a new Git repository.
- Familiarize yourself with basic Git commands such as git add, git commit, and git status to track changes and manage your repository.
- Learn the concepts of branching and merging in Git to manage different versions of your code and collaborate with team members effectively.
- Understand the concept of remote repositories and learn how to work with remote repositories in Git for collaboration and backup purposes.
- Learn how to collaborate with team members using Git, including pushing and pulling changes to and from remote repositories, resolving conflicts, and reviewing code.

Day 31

GIT

Key Topics: Introduction to Version Control, Understanding Git, Setting Up Git, Basic Git Commands, Branching and Merging, Remote Repositories, Collaborating with Git

Continuous Learning: Technical Enablement

Git Complete: The definitive, step-by-step guide to Git

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.

- **Section 1:** Introduction
- **Section 2:** Installation
- **Section 3:** Git Quick Start
- **Section 4:** GitHub Updates
- **Section 6:** Basic Git Commands
- **Section 8:** Comparisons
- **Section 9:** Branching and Merging

- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Git Config
- Clone Repo
- Add, Commit And Push
- Pull And Merge
- Merge - Resolve Conflict
- Git Tags

Stage 3 -> Course 2 -> Application Debugging - Backend

Course Overview

In **Course 2** of **Stage 3**, you will delve into the essential skill of **debugging Spring Boot applications**, particularly focusing on REST APIs. Debugging is a critical aspect of software development, allowing you to identify and fix issues in your code efficiently. You will learn how to set up debugging in a Spring Boot application using popular IDEs like **IntelliJ IDEA** or **Eclipse**, enabling you to step through your code, inspect variables, and understand the flow of execution.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand the role of debugging in improving code quality.
- Set up debugging in Spring Boot using IntelliJ IDEA or Eclipse.
- Use breakpoints to pause and control code execution.
- Step through code to analyze method calls and flow.
- Inspect and modify variable states during execution.
- Debug REST API endpoints and HTTP requests.
- Capture and diagnose exceptions and errors.
- Apply logging techniques alongside debugging.
- Develop a systematic approach to debugging.

Day 32 - Forenoon

Application Debugging - Backend

Key Topics: Debugging Fundamentals, Debugging REST Endpoints, Debugging Asynchronous Code

Demo Videos

Watch the following demo videos provided in the platform for getting a complete picture about how to debug backend REST API using Eclipse.

Note: The below given Cognizant Kpoint videos can be accessed even after your Tekstac platform access is revoked.

- Debugging REST API-Part1
- Debugging REST API-LogFiles

Additional Learning

Eclipse Debugging Techniques And Tricks

- Go through the entire course.

Click on the links provided and delve into the content to enhance your understanding.

- Debugging Spring Applications
- Spring Boot – @Async Annotation

Day 32 - Afternoon, 33 - Forenoon

- These parts of the days have been included in adjusting the behavioral training and mentor connect.

Stage 3 -> Course 3 -> Spring Data JPA with Spring Boot, Hibernate

Course Overview

In **Course 3** of **Stage 3**, you will delve into **Spring Data JPA** integration with **Spring Boot 3**, leveraging **Hibernate** as the JPA provider to streamline data access layer development in Spring applications. This integration simplifies database interactions by abstracting underlying data access technologies and providing a repository-based programming model. You will explore entity mapping using JPA annotations, repository interfaces for CRUD operations, custom query methods for data retrieval, derived queries from method names, and features like pagination, sorting, auditing, and transaction management. By the end of this milestone, you will have a comprehensive understanding of how to harness the power of Spring Data JPA with Spring Boot 3 and Hibernate to build efficient and robust data access layers for your applications.

Learning Objectives

After completing this course, GenCs will be able to:

- Explain the role of Hibernate as a JPA provider in Spring applications.
- Understand the benefits of using Spring Data JPA to abstract data access logic.
- Map entity relationships such as One-to-One, One-to-Many, and Many-to-Many.
- Perform CRUD operations (Create, Read, Update, Delete) using repositories.
- Write native and JPQL queries to retrieve specific data sets.
- Set up and configure auditing features to track entity creation and updates.
- Understand the importance of transaction management in data operations.
- Configure and manage transactions using @Transactional annotations.
- Handle database-related exceptions gracefully using Spring mechanisms.
- Write unit tests for repositories using in-memory databases (H2, HSQL).
- Optimize query performance with caching and batch fetching techniques.

Day 33 - Afternoon

- This part of the day has been included in adjusting the behavioral training and mentor connect.

Day 34

Spring Data JPA with Spring Boot, Hibernate

Key Topics: Introduction to Spring Data JPA, Setting Up a Spring Boot Project with Spring Data JPA

Continuous Learning: Technical Enablement

Hibernate and Spring Data JPA: Beginner to Guru

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 2:** Introduction to Spring Data JPA
    - **Section 5:** Hibernate with MySQL
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Offline Hands-On (Additional Practice)

- Try out the offline hands-on exercises given in the learning path on **Spring Data JPA**.

Day 35

Spring Data JPA with Spring Boot, Hibernate

Key Topics: Entity Mapping, Spring Data Repositories, CRUD Operations with Spring Data JPA

Continuous Learning: Technical Enablement

Hibernate and Spring Data JPA: Beginner to Guru

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 16:** Database Relationship Mapping
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Click on the link provided and delve into the content to enhance your understanding.

- JPA Repositories
- Spring Boot JPA CRUD with Example

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- School Strength
- Loan Lenders
- Sales Statistics
- Account with ATM Card With Logger

Day 36

Spring Data JPA with Spring Boot, Hibernate

Key Topics: Query Methods and Named Queries, Pagination and Sorting, Auditing with Spring Data JPA, Spring Data JPA Projections

Continuous Learning: Technical Enablement

Hibernate and Spring Data JPA: Beginner to Guru

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 13:** Spring Data JPA Queries
    - **Section 14:** Paging and Sorting

- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Click on the link provided and delve into the content to enhance your understanding.

- Auditing with JPA, Hibernate, and Spring Data JPA
- Spring Data JPA Projections

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Account Transaction

Day 37

Spring Data JPA with Spring Boot, Hibernate

Key Topics: Spring Data JPA and Spring Boot Integration, Spring Data JPA and Hibernate

Continuous Learning: Technical Enablement

Click on the link provided and delve into the content to enhance your understanding.

- How to use Spring Data JPA in Spring Boot Project
- Spring Data JPA vs Hibernate

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Country - State(OneToMany Bi-Directional)
- Component Mapping

Code Challenges (For Practice Only)

Attempt the following Code Challenges through the Learning Path for checking your skill level on user Spring Data JPA. There will be only 3 attempts and you have to score 70% in order to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - Spring Data JPA with Spring Boot

Offline Hands-On (Additional Practice)

- Try out the offline hands-on exercises given in the learning path on **Spring Data JPA**.

Stage 3 -> Course 4 -> Spring REST using Spring Boot 3

Course Overview

Course 4 of **Stage 3** covers **Spring REST using Spring Boot 3**. In this milestone, participants will learn about building RESTful web services using the Spring Framework, specifically focusing on the latest version, Spring Boot 3. Spring Boot is a popular framework for rapidly developing and deploying Java-based web applications.

Learning Objectives

After completing this course, GenCs will be able to:

- Grasp the fundamentals of REST, including statelessness, resource-based interactions, and HTTP methods (GET, POST, PUT, DELETE).
- Set up a Spring Boot project and build REST controllers to handle HTTP requests and return JSON responses.
- Manage path variables, query parameters, request bodies, response headers, and implement exception handling using Spring annotations.
- Use strategies like path/version headers and content negotiation to manage API versions and support multiple media types (e.g., JSON, XML).
- Implement authentication and authorization, including JWT-based token authentication, and manage CORS policies.
- Apply DTOs to decouple data models from entities and customize JSON serialization/deserialization.

- Integrate Actuator to expose metrics and health endpoints, and configure custom monitoring and security.
- Write unit and integration tests with JUnit, Mockito, and MockMvc, and generate API documentation using Swagger/OpenAPI.

Day 38

Spring REST using Spring Boot 3

Key Topics: Introduction to Spring REST and Spring Boot 3, Building a Simple REST Controller

Continuous Learning: Technical Enablement

Building Real-Time REST APIs with Spring Boot - Blog App

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 3:** REST Basics and Key Concepts (For Beginners)
  - **Section 4:** Spring Boot REST API Development Basics - Important Annotations
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Restaurant Menu Service
- Movie Ticket Booking REST Controller App

Day 39

Spring REST using Spring Boot 3

Key Topics: Request and Response Handling, RESTful Resource Representation with DTOs

Continuous Learning: Technical Enablement

Building Real-Time REST APIs with Spring Boot - Blog App

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 6:** Building CRUD REST API's for Post Resource
  - **Section 9:** Using ModelMapper-Map Entity to DTO and Vice Versa
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Online Event Management - Get and Post
- Online Event Management - Put and Delete
- Patient Management - Exception Handling
- Inventory Management System - Get and Post
- Inventory Management System - Put and Delete
- Online Recipe Management - Exception Handling

Day 40

Spring REST using Spring Boot 3

Key Topics: RESTful HATEOAS, Content Negotiation and Media Types

Continuous Learning: Technical Enablement

Click on the link provided and delve into the content to enhance your understanding.

- Building a Hypermedia-Driven RESTful Web Service
- Spring Boot REST Web Services Content Negotiation

Day 41

Spring REST using Spring Boot 3

Key Topics: Spring Boot Actuator for REST Monitoring, Security and Authentication in RESTful APIs

Continuous Learning: Technical Enablement

Click on the link provided and delve into the content to enhance your understanding.

- Spring Boot Actuator
- Authentication and Authorization in Spring Boot 3.0 with Spring Security

Day 42

Spring REST using Spring Boot 3

Key Topics: Testing RESTful APIs, Documenting RESTful APIs

Continuous Learning: Technical Enablement

Building Real-Time REST APIs with Spring Boot - Blog App

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 18:** REST API Documentation using SpringDoc OpenAPI in Spring Boot 3
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Click on the link provided and delve into the content to enhance your understanding.

- Spring Boot Unit Testing REST APIs Tutorial

Code Challenges (For Practice Only)

Attempt the following Code Challenges through the Learning Path for checking your skill level on Spring REST using Spring Boot. There will be only 3 attempts and you have to score 70% in order to clear this challenge.

> Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - Spring REST using Spring Boot

IDP - Project Activities

Day 43

Internal Demo and Rework

- This day can be utilized for Internal Demo & Re-work

Stage 3 -> Course 5 -> Microservices with Spring Boot 3 and Spring Cloud

Course Overview

Course 5 of **Stage 3** covers **Microservices with Spring Boot 3 and Spring Cloud**. This milestone focuses on introducing participants to building microservices-based applications using Spring Boot 3 and Spring Cloud. It covers fundamental concepts of microservices architecture, including modularity, scalability, resilience, and decentralized data management. Participants will learn how Spring Boot 3 simplifies the development of microservices with its auto-configuration and opinionated defaults. Additionally, they will explore how Spring Cloud complements Spring Boot by providing tools for building distributed systems, such as service discovery, circuit breakers, and distributed tracing. The milestone includes practical examples and hands-on exercises to reinforce learning.

Learning Objectives

After completing this course, GenCs will be able to:

- Identify the characteristics, advantages, challenges, and appropriate use cases for microservices.
- Set up service discovery and registration with Netflix Eureka.
- Implement authentication and authorization for RESTful APIs in a microservices environment.
- Configure centralized authentication using OAuth 2.1, OIDC, and JWT-based secure communication.
- Use Spring Cloud Feign for declarative REST clients and explore service orchestration patterns.
- Implement circuit breakers and fallback mechanisms with Spring Cloud Circuit Breaker for fault tolerance.
- Set up API Gateway with Spring Cloud Gateway for routing, filtering, and load balancing.
- Manage dynamic and externalized configurations using Spring Cloud Config.
- Use Spring Boot Actuator for endpoint monitoring and integrate tools like Prometheus and Grafana for metrics.
- Implement role-based access control (RBAC), secure inter-service communication, and enforce security policies.

Day 44

Microservices with Spring Boot 3 and Spring Cloud

Key Topics: Introduction to Microservices Architecture (MSA), Spring Cloud for Microservices

Continuous Learning: Technical Enablement

[NEW] Building Microservices with Spring Boot & Spring Cloud

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 10:** Microservices Introduction
    - **Section 11:** Building Microservices
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 45

Microservices with Spring Boot 3 and Spring Cloud

Key Topics: Microservices Communication with Spring Cloud, API Gateway and Edge Services

Continuous Learning: Technical Enablement

[NEW] Building Microservices with Spring Boot & Spring Cloud

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 13:** Microservices Communication
    - **Section 15:** API Gateway using Spring Cloud Gateway
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Register Order App in Spring Cloud
- Access Order App Via Router
- Inter- Microservice Communication UserApp and ProductApp
- Service Fallback

Day 46

Microservices with Spring Boot 3 and Spring Cloud

Key Topics: Spring Cloud Config

Continuous Learning: Technical Enablement

[NEW] Building Microservices with Spring Boot & Spring Cloud

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 16:** Centralized Configurations using Spring Cloud Config Server
    - **Section 17:** Auto Refresh Config Changes using Spring Cloud Bus
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- E-Commerce Platform

Day 47

Microservices with Spring Boot 3 and Spring Cloud

Key Topics: Monitoring and Metrics in Microservices

Continuous Learning: Technical Enablement

Click on the link provided and delve into the content to enhance your understanding.

- Monitoring Spring Boot Microservices with Prometheus
- Monitoring Made Simple: Empowering Spring Boot Applications with Prometheus and Grafana

Day 48 - Forenoon

Microservices with Spring Boot 3 and Spring Cloud

Key Topics: Security Best Practices in Microservices

Continuous Learning: Technical Enablement

Click on the link provided and delve into the content to enhance your understanding.

- Role-Based Access Control with Spring Security
- Learn How to Secure Service-to-Service Microservices

Code Challenges (For Practice Only)

Attempt the following Code Challenges through the Learning Path for checking your skill level on Microservices. There will be only 3 attempts, and you must score 70% to clear this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - Microservices

Day 48 - Afternoon, 49, 50, 51 - Forenoon

Microservices Implementation

- These days will be spent on the following IDP development activity.
  - Implement Microservices using Spring Cloud

Day 51 - Afternoon

- This part of the day has been included in adjusting the behavioral training and mentor connect.

Interim Evaluation (Project + Technical)

Day 52, 53

Interim Evaluation (Project + Technical)

- Interim evaluation will be conducted on these days, and the mode will be a video interview on the Tekstac platform. There will be ONLY one attempt.

Stage 4 - Modern Frontend Development and Cloud Technologies

Overview

Stage 4 deals with the Niche skills such as Angular, Docker, Cloud and DevOps basics using GCP which are seminal while developing/maintaining various software applications. We provide unique learning experience to learners by including diversified learning content and learning methodologies that are based on adult learning principles.

As part of **Stage 4** of your training, the following skills will be covered.

- Docker basics
- Cloud and DevOps Basics using GCP
- Angular (v16)
- Application Debugging - Front-end

How and From Where to Learn?

- You can learn from the sources as they are mentioned in this learning guide.

Stage 4 -> Course 1 -> Docker Basics

Course Overview

Course 1 of **Stage 4** introduces learners to **Docker**; a popular containerization platform used in modern application development. Docker allows developers to package applications and their dependencies into containers, which can run consistently across different environments. The module covers key concepts such as Docker containers, images, volumes, and networking. Learners will understand the benefits of Docker, including improved portability, scalability, and efficiency in resource utilization. They will also learn how to create, manage, and deploy Docker containers, making it easier to build, ship, and run applications in a consistent and isolated environment.

Learning Objectives

After completing this course, GenCs will be able to:

- Understand Containerization Concepts and Docker Architecture
- Develop Proficiency in Working with Docker
- Explore Dockerfile and Image Building
- Manage Container Communication and Persistence
- Use Docker Compose for Multi-Container Applications
- Integrate Docker with REST API and Databases

Day 54

Docker Basics

Key Topics: Introduction, Docker Commands. Docker Run, Docker Images, Docker Compose

Continuous Learning: Technical Enablement

Docker for the Absolute Beginner - Hands On - DevOps

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 1:** Introduction

- Section 2: Docker Commands
- Section 3: Docker Run
- Section 4: Docker Images

- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 55

Docker Basics

Key Topics: Docker Registry, Docker Engine, Docker Storage, Docker Networking, Container Orchestration

Continuous Learning: Technical Enablement

Docker for the Absolute Beginner - Hands On - DevOps

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 6:** Docker Registry
    - **Section 7:** Docker Engine, Storage and Networking
    - **Section 9:** Container Orchestration - Docker Swarm & Kubernetes
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Stage 4 -> Course 2 -> Cloud and DevOps

Course Overview

Course 2 of **Stage 4** is designed to provide learners with a comprehensive understanding of cloud computing concepts and their application through **GCP**. The course introduces foundational topics, such as cloud computing models, deployment strategies, and service providers, with a deep dive into GCP's infrastructure, services, and tools. Students will explore core GCP services, including compute, storage, networking, databases, and identity management, along with serverless computing and **DevOps** practices. By the end of the course, learners will be equipped to leverage GCP effectively for modern cloud-based solutions and implement DevOps pipelines for streamlined deployment.

Learning Objectives

After completing this course, GenCs will be able to:

- Explain the principles of cloud computing, including its service and deployment models.
- Identify key advantages of cloud computing and the diverse services offered by major providers, with a focus on GCP.
- Describe and utilize GCP's compute, storage, networking, and database services for various use cases.
- Implement Identity and Access Management (IAM) to ensure secure resource management.

- Understand and utilize serverless solutions available in GCP for efficient application deployment.
- Gain proficiency in GCP DevOps tools, including Cloud Source Repositories, Cloud Build, Cloud Deploy, and Build Pipelines.
- Design and implement CI/CD pipelines for automated and streamlined application deployment.
- Apply theoretical knowledge to hands-on projects involving GCP services and DevOps pipelines.
- Solve real-world problems by designing and deploying robust cloud-based solutions.

Day 56

Cloud and DevOps Basics using GCP

Key Topics: Introduction to Cloud Computing, Cloud Service Models, Cloud Deployment Models, Cloud Service Providers, Advantages of Cloud Computing and Various Services Provided by GCP, Introduction to GCP, GCP Compute Services

Continuous Learning: Technical Enablement

A Practical Introduction to Cloud Computing

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope. course
    - **Section 2:** A (Very) Brief History of Traditional IT Deployments
    - **Section 3:** Defining Cloud - Essential Characteristics
    - **Section 4:** Cloud Service Models - IaaS, PaaS and SaaS
    - **Section 5:** Cloud Deployment Models - Public, Private and Hybrid Cloud
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

GCP for Beginners - Become a Google Cloud Digital Leader

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope. course
    - **Section 1:** Getting Started - Google Cloud - Cloud Digital Leader Certification
    - **Section 2:** Regions and Zones in GCP - Google Cloud Platform
    - **Section 3:** Getting started with Compute Engine, Instance Groups & Load Balancing
    - **Section 4:** Managed Services - IAAS, PAAS and SAAS in Google Cloud Platform
    - **Section 5:** Exploring Google Cloud Platform GCP Compute Services
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 57

Cloud and DevOps Basics using AWS

Key Topics: GCP Storage Services, GCP Networking, GCP Database Services, GCP Identity and Access Management (IAM), GCP Serverless Computing

Continuous Learning: Technical Enablement

GCP for Beginners - Become a Google Cloud Digital Leader

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 6:** Block, File and Object Storage in GCP
  - **Section 7:** Databases in Google Cloud
  - **Section 9:** Getting started with Cloud IAM
  - **Section 18:** Exploring Cloud Native and Other Modern Architectures
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 58

Cloud and DevOps Basics using AWS

Key Topics: Introduction to DevOps in GCP, GCP Cloud Source Repositories, GCP Cloud Build, GCP Cloud Deploy, GCP Cloud Build Pipelines

Continuous Learning: Technical Enablement

Google Cloud Developer - GCP Professional Certification

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **Section 15:** Exploring DevOps and SRE
  - **Section 16:** Getting Started with Cloud Build
  - **Section 30:** Release Management in Google Cloud
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Stage 4 -> Course 3 -> Angular (v16)

Course Overview

Course 3 of **Stage 4** provides a comprehensive introduction to building modern, dynamic web applications using **Angular** and **Node.js**. Designed for aspiring web developers, the curriculum covers foundational concepts and advanced topics, empowering learners to create robust and scalable applications. The course emphasizes TypeScript, Angular components, directives, forms, state management, and routing, as well as server-side programming with Node.js. Learners will also explore reactive programming with RxJS and gain hands-on experience with API integrations, testing strategies, and asynchronous programming. By the end of the course, participants will have a holistic understanding of front-end and back-end development using Angular and Node.js.

Learning Objectives

After completing this course, GenCs will be able to:

- Set up and configure the Angular development environment for building web applications.
- Master TypeScript essentials to write and manage Angular code efficiently.
- Develop and utilize Angular components, directives, and pipes to create modular and reusable UI elements.
- Implement Angular forms with validation for capturing and managing user input.
- Leverage dependency injection and services for efficient data sharing and application logic organization.
- Configure and use Angular routing and navigation to enable seamless user experiences in multi-page applications.
- Explore advanced router features for managing application flow and guarding routes.
- Integrate HTTP client functionality to interact with RESTful APIs and handle asynchronous data.
- Understand and apply state management principles for managing complex application states.
- Adopt reactive programming techniques with RxJS for managing asynchronous data streams.
- Write and execute tests to ensure the reliability and quality of Angular applications.
- Gain a foundational understanding of Node.js for server-side programming.
- Work with core Node.js modules to handle file systems, events, and streams.
- Understand asynchronous programming in Node.js to manage non-blocking operations effectively.

Day 59

Angular (v16)

Key Topics: Introduction to Angular and Setting Up Environment, Angular Basics

Continuous Learning: Technical Enablement

Angular - The Complete Guide (2024 Edition)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope. course
    - **Section 1:** Getting started
    - **Section 18:** The Basics [Angular < 16]
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Day 60

Angular (v16)

Key Topics: TypeScript Essentials for Angular

Continuous Learning: Technical Enablement

Angular - The Complete Guide (2024 Edition)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 46:** Bonus: TypeScript Introduction (For Angular 2 Usage)
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Offline Hands-On (Additional Practice)

- Try out the offline hands-on exercises given in the learning path on **Angular**.

Day 61

Angular (v16)

Key Topics: Angular Components, Directives and Pipes

Continuous Learning: Technical Enablement

Angular - The Complete Guide (2024 Edition)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 21:** Components & Databinding Deep Dive [Angular < 16]
    - **Section 23:** Directives Deep Dive [Angular < 16]
    - **Section 33:** Using Pipes to Transform Output [Angular < 16]
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

> Do not copy paste the code. Write the code yourself.

- My First - Sports Club App
- Boat Ride Entrance Fee
- Show Student DB - ngFor Directive
- Remove Student - (component, ngFor)
- Flight Reservation Form - (Dynamic dropdown using ngFor)
- Change Cases – Pipe
- Book House - Cross component communication

Offline Hands-On (Additional Practice)

- Try out the offline hands-on exercises given in the learning path on **Angular**.

Day 62

Angular (v16)

Key Topics: Angular Forms, Dependency Injection and Services

Continuous Learning: Technical Enablement

Angular - The Complete Guide (2024 Edition)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope. course
  - **Section 31:** Handling Forms in Angular Apps [Angular < 16]
  - **Section 25:** Using Services & Dependency Injection [Angular < 16]
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Offline Hands-On (Additional Practice)

- Try out the offline hands-on exercises given in the learning path on **Angular**.

Day 63

Angular (v16)

Key Topics: Routing and Navigation, Router Features

Continuous Learning: Technical Enablement

Angular - The Complete Guide (2024 Edition)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 27:** Changing Pages with Routing [Angular < 16]
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Hands-On

Complete the following set of hands-on given in the Learning Path at Tekstac.

Do not copy paste the code. Write the code yourself.

- Rent A Home - Routing with Query param

Offline Hands-On (Additional Practice)

- Try out the offline hands-on exercises given in the learning path on **Angular**.

Code Challenge (For Practice Only)

Take on the following Code Challenge in the Tekstac Learning Path to assess your skill level in Angular. You will have 3 attempts, and you must score 70% or higher to pass this challenge.

Do not copy paste the code. Write the code yourself.

- Assess-Type-1: Code Challenge - Angular

Day 64

Angular (v16)

Key Topics: HTTP Client and APIs, State Management in Angular

Continuous Learning: Technical Enablement

Angular - The Complete Guide (2024 Edition)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 34:** Making Http Requests [Angular < 16]
    - **Section 41:** Bonus: Using NgRx For State Management
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Offline Hands-On (Additional Practice)

- Try out the offline hands-on exercises given in the learning path on **Angular**.

Day 65

Angular (v16)

Key Topics: Reactive Programming with RxJS

Continuous Learning: Technical Enablement

Angular - The Complete Guide (2024 Edition)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 29:** Understanding Observables [Angular < 16]
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Offline Hands-On (Additional Practice)

- Try out the offline hands-on exercises given in the learning path on **Angular**.

Day 66

Angular (v16)

Key Topics: Testing Angular Applications

Continuous Learning: Technical Enablement

Angular Testing Masterclass (Angular 19)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **ALL** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Offline Hands-On (Additional Practice)

- Try out the offline hands-on exercises given in the learning path on **Angular**.

Day 67

Angular (v16)

Key Topics: Introduction to Node.js, Core Node.js Modules, Asynchronous Programming Basics

Continuous Learning: Technical Enablement

The Complete Node.js Developer Course (3rd Edition)

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
    - **Section 2:** Installing and Exploring Node.js
    - **Section 3:** Node.js Module System (Notes App)
    - **Section 4:** File System and Command Line Args (Notes App)
    - **Section 6:** Asynchronous Node.js (Weather App)
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Stage 4 -> Course 4 -> Application Debugging - Front-end

Course Overview

Course 4 of **Stage 4** training focuses on front-end application debugging in Angular. In this module, learners will gain essential skills in identifying and resolving issues in their front-end applications. They will learn debugging techniques specific to Angular, including using developer tools, logging, and debugging libraries. The module will cover common debugging scenarios such as troubleshooting errors, optimizing performance, and understanding application behavior.

Learning Objectives

After completing this course, GenCs will be able to:

- Familiarize with Chrome DevTools and VS Code Debugger for web development.
- Configure Source Maps and adjust project settings for effective debugging.
- Inspect state, props, and services using breakpoints.
- Debug Observables, Promises, and Async/Await to resolve async issues.
- Monitor Angular Route Guards for smooth navigation.

- Track network requests and profile performance to optimize applications.
- Use memory tools to detect leaks and improve application stability.
- Debug across multiple browsers to handle browser-specific issues.

Day 68 - Forenoon

Application Debugging - Front-end

Key Topics: Debugging Angular Applications with Chrome DevTools

Continuous Learning: Technical Enablement

Devtools Pro: The Basics of Chrome Developer Tools

- Walkthrough the following Udemy course sections and focus on the corresponding topics within our training curriculum's technical scope.
  - **All** Sections
- Ensure that you learn these topics through self-learning and practice alongside the course instructor. It is NOT necessary to cover every topic comprehensively within each section.

Demo Videos

Watch the following demo videos provided in the platform to get a complete picture about how to debug an Angular application using Chrome's DevTools and VS Code Debugger.

Note: The below given Cognizant Kpoint videos can be accessed even after your Tekstac platform access is revoked.

- Debugging Angular Part 1
- Debugging Angular Part 2

IDP - Project Activities

Day 68 - Afternoon, 69, 70 - Forenoon

Integration

These days will be spent on the following IDP activity with the Trainer guidance.
- Integration of Front-end with Web API

Final Evaluation (Project + Technical)

Day 70 - Afternoon, 71, 72, 73 - Forenoon

Final Evaluation (Project + Technical)

- Final Evaluation will be conducted on these days, and the mode will be a video interview on the Tekstac platform.

Final Assessment

Day 73 - Afternoon, 74 - Forenoon

HackerRank Prep & Assessment

- Learners will have HackerRank assessment on this day.

How to learn each day?

Each day has a set of learning objectives. These learning objectives can be met by going through the Udemy courses and by completing the hands-on exercises mentioned in the daily plan.

The strategies below will help you decide the learning approach.

Learning Strategy & Approach

Find below few imaginary profiles. For each of these profiles we have defined a recommended learning approach. This is not an exhaustive list. The approaches below might help invent a new way of learning.

Profile #1

	Harry Reacher **Engineering Discipline:** Electronics **Skills:** Python, Ruby on Rails, nginx **Project:** Mining Crime Data to get Route Cause Insights

Learning Approach to Programming Languages: I do not want to waste my time learning. I am more practice oriented. I want to work on the problem immediately

What will work for me?

- Directly complete hands-on exercises
- Refer Internet or Udemy Courses
- If hands on are implemented early, clarify your friends' questions and troubleshoot their issues

Profile #2

Olivia Richards

Engineering Discipline: Computer Science

Skills: Java, C, C++

Project: Library Management System

Learning Approach to Programming Languages: I have interest, but I don't know where to start.

What will work for me?

- Go through the recommended Udemy Course
- Try completing the hands-on exercises
- Get your clarifications solved with help from Tech SME
- Get help from other learners in your batch whom had already completed

Profile #3

Greg Anderson

Engineering Discipline: Civil

Skills: C

Project: Fiber reinforced concrete

Learning Approach to Programming Languages: I am scared of programming languages. I haven't got my hands dirty with coding

What will work for me?

- Go through the recommended Udemy Course
- Implement the coding along with the author of the Udemy Course
- Try completing the hands-on exercises
- Clarify queries with SME
- Troubleshoot programming issues with help from SME or learner from your classroom whom had already completed

FAQs

1. **Who can participate in this program?**

   Students who have enrolled for Full Internship Program (or) the Cognizant on-boarded GEN Cs can participate in this program.

2. **Is there any pre-learning I should do?**

   No. This program is open to all students from any academic discipline.

3. **What is Code Challenge?**

   A problem statement will be provided to you, and you need to solve it using a single skill.

4. **What is Integrated Capability Test (ICT)?**

   A case study problem statement will be provided to you, that you may need solve using the combination of Skills learnt in the given stage.

5. **How many attempts are provided for the Coding challenge and ICTs? Is it open all the time for practice?**

   The Coding challenges and ICTs are open and there are 3 attempts to take them up.

6. **What are the entry criteria for Stage 1 Assessment?**

   A 100% hands-on completion and attempt in CC & ICT are the eligibility criteria for the Stage 1 Assessment.

7. **What skills are covered in the qualifier?**

   The skills of Stage 1 are covered in the qualifier. Only ONE attempt is provided to clear with a minimum score of 70%

8. **What if I fail in the Interim evaluation?**

   Your coach will notify your performance in the Interim evaluation. However, you can continue with the learning.

9. **How many chances will I get in the Final evaluation?**

   You'll get 2 chances in the Final evaluation which covers ALL the skills in the learning journey.

10. **Will we be provided with Projects to work on?**

    No, you will have to ideate, design and develop the project which will be reviewed and assessed by the project mentor.

11. **Whom do I reach out in case of any queries?**

    Batch Owner is your point of contact.